

From CAIG Decline to a Fundable Program

Diagnosis, technical upgrades, and a six-month resubmission roadmap for the Central Indiana mixed-autonomy traffic digital twin

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For discussion with **Prof. Mohammad Al Hasan** and **Prof. Aniruddha Banerjee** · Based on the CAIG proposal “GeoAI Traffic Twin” and the program landscape as verified on August 13, 2026

Executive summary

Why the proposal was declined, in one sentence. CAIG (NSF 25-530) funds AI that advances scientific understanding of the *Earth system* — the research portfolio of NSF’s Geosciences Directorate — and a transportation digital twin, however geospatially sophisticated, answers transportation-engineering questions rather than Earth-system questions. In NSF’s taxonomy, GIS is a data infrastructure, not a geoscience. The proposal’s own central fit argument (“the usage of GIS as the core integrative layer makes this CAIG project genuinely co-equal in AI and Geosciences”) is precisely the claim a GEO-led panel is trained to reject, which is fully consistent with a “not a good fit” decline.

The strategic good news. Nothing about the decline impugns the underlying assets. The INDOT partnership, the ATLAS 11-county lane-level network, two decades of TransModeler depth, and the complexity-science framing are genuinely differentiating — they were simply aimed at the wrong directorate. Repositioned as a *learning-enabled cyber-physical system under environmental stress*, the same project sits squarely inside NSF’s new CISE **Future CoRe** solicitation (NSF 25-543, which absorbed the former CPS and Smart & Connected Communities programs; up to \$1M over up to 4 years; proposals accepted anytime with target dates of **September 10, 2026** and **February 4, 2027**), and inside FHWA’s FY2026 Exploratory Advanced Research interest in data science for highway transportation.

What I propose to contribute. I would join as co-PI and own a *Learning-Accelerated Simulation and Safe Multi-Agent Learning* thrust, structured as five work packages: (WP1) graph world-model surrogates of the microsimulator; (WP2) simulation-based-inference calibration with uncertainty quantification; (WP3) safety-shielded cooperative multi-agent RL; (WP4) a learned emulator of percolation and cascade metrics; and (WP5) generative stress-scenario synthesis. These directly repair the four vulnerabilities most likely to sink a CISE resubmission — compute feasibility, validation, uncertainty, and safety — while adding the publishable AI novelty the current draft under-claims. Everything in WP1–WP5 runs on the standard deep-learning/RL stack I use daily; no exotic hardware, no dependence on my prior neuromorphic tooling.

Recommended path. Target the Future CoRe CPS program at the **February 4, 2027** panel cycle (September 10 is available but leaves no room for pilot results); run a 10-week pilot this fall so the resubmission ships with preliminary evidence; and in parallel open two USDOT doors — the FHWA EAR FY2026 data-science solicitation and an INDOT-led SMART Stage 1 application.

Five verified facts that shape the strategy (details and links in §6 and Appendix B)

1. CAIG’s scope is “scientific understanding of the Earth system,” with mandatory Geosciences Advancement / AI Advancement / Partnership subsections; its last posted deadline was Feb 4, 2026 and no next cycle is currently announced.
2. NSF consolidated the CISE core, CPS, and S&CC solicitations into Future CoRe (NSF 25-543): one project class up to \$1M / 4 years, accepted anytime, target dates the 2nd Thursday of September and 1st Thursday of February.

3. Future CoRe caps each individual at two proposals as PI/co-PI/Senior Personnel per rolling 12 months across all its programs.
4. FHWA's EAR program posted an FY2026 solicitation focused on data-science applications in highway transportation (verify current dates on SAM.gov).
5. IU Indianapolis earned Carnegie R1 status in Feb 2025 — which, notably, makes us *ineligible* for CRII (it excludes R1-affiliated PIs), so the team resubmission is also my own fastest funding path.

1. What the proposal proposes (faithful one-paragraph-per-thrust summary)

The proposal builds a geospatially faithful digital twin of Central Indiana's road network — Marion County plus Hendricks, Hamilton, Johnson, Boone, and Morgan; I-65/I-70, arterials, collectors, and rural connectors — inside the TransModeler microsimulation platform, fusing ATLAS street centerlines, USGS LiDAR-derived grades and sight distances, INDOT real-time and historical traffic data, and ten years of snow/precipitation records. **Thrust I** constructs and calibrates the twin (Gaussian-process surrogate calibration against a 0%-CAV baseline) and implements V2V/V2I-enabled route guidance, freeway speed harmonization, and traffic-light optimal speed advisory. **Thrust II** populates the twin with three vehicle classes — calibrated human drivers (IDM/MOBIL/gap acceptance), rule-based SAE L2–3 vehicles, and SAE L4–5 CAVs controlled by multi-agent PPO plus Stackelberg / Nash-bargaining modules at merges and intersections — to test whether cooperative multi-objective learning beats selfish optimization on congestion and safety. **Thrust III** applies complexity science — phase transitions in network order parameters, percolation thresholds of the CAV communication graph, cascade dynamics, and Kuramoto-style synchronization — to locate critical penetration levels and vulnerable corridors, ending in a predictive framework and agency-facing tools (CAV Policy Evaluator dashboard, resilience maps) delivered with INDOT and IGIO. Two PhD students, two new courses, and annual practitioner workshops round out the plan. This is a strong engineering core with real institutional assets; the sections below are about packaging, rigor, and venue, not about the vision.

2. Why CAIG said “not a good fit” — and what that implies

CAIG is run out of the Geosciences Directorate (with CISE and MPS as partners) and its synopsis is explicit: the program “seeks to advance the development and adoption of innovative AI methods to increase scientific understanding of the Earth system.” Every proposal must argue three required subsections — *Geosciences Advancement*, *AI Advancement*, and *Partnership* — and the geoscience half must be a discovery contribution to atmospheric, ocean, Earth, or geospace science, evaluated by geoscientists. Read against that bar, three things doomed us regardless of technical quality:

- **The “Geo” in our GeoAI is geospatial, not geoscience.** Lane-level GIS, LiDAR grades, and land-use layers are data engineering for a transportation model. No Earth-system question is answered anywhere in the proposal; weather enters only as a forcing input to traffic, never as an object of scientific study.
- **No geoscientist among senior personnel.** The team is ML + GIS/transportation; the evaluators (IGIO) are GIS practitioners. A CAIG panel looks for an atmospheric/hydrologic/earth scientist with co-equal intellectual ownership and finds none.
- **The fit argument invited the rejection.** Asserting that GIS-as-integrative-layer makes the project “co-equal in AI and Geosciences” reads to a GEO panel as a category error, and likely triaged the proposal early — consistent with a decline on fit rather than a science critique.

Implications. (a) Do not resubmit to CAIG in its current form. (b) A CAIG-compliant version is a *different project*: e.g., AI for mesoscale winter precipitation and road-surface state estimation (RWIS, HRRR, radar assimilation) with the traffic twin demoted to an impact layer — viable only with an atmospheric-science co-PI, and moot for now since no next CAIG deadline is posted. (c) The project's natural reviewers live in CISE (cyber-physical systems, multi-agent learning) and at USDOT/FHWA — so that is where we go.

3. Ten weaknesses a CISE panel would still flag — each with its fix

Fixing the venue is necessary but not sufficient. Below is the red-team read I would give any colleague, ordered by how likely each issue is to draw a fatal reviewer comment at CPS/Robust-Intelligence panels. Each fix points into the work packages of §5 or the cross-cutting plan of §5.6.

W1. Compute feasibility of MARL at metro scale is unaddressed.

Thrust II promises MAPPO training over “tens of millions of steps” with thousands of agents inside a commercial microsimulator, but names no compute resources and no acceleration strategy. Reviewers will estimate wall-clock and stop reading. **Fix:** WP1 world-model surrogate (target: two orders of magnitude faster policy training) plus a named facilities plan — IU’s Big Red 200 GPUs, the Jetstream2 NSF cloud housed at IU for elastic CPU simulation farms, and a NAIRR Pilot allocation request.

W2. The validation plan cannot support the headline claims.

The twin is calibrated only at 0% CAV penetration, then the “reality gap” is “extrapolated” as penetration grows — an extrapolation with no ground truth and no external anchor. **Fix:** a three-tier validation ladder (§5.6): held-out-day replication at 0% penetration; reproduction of established mixed-autonomy results (e.g., the Stern et al. ring-road field experiment and the Flow benchmark findings the draft already cites) as go/no-go gates; and perception/communication degradation priors drawn from public winter AV datasets rather than assumed.

W3. Calibration is point-estimate and dated; uncertainty never reaches the conclusions.

GP-based surrogate calibration yields point estimates for dozens of context-dependent parameters, with identifiability unexamined and no uncertainty propagated into policy conclusions — risky when the deliverable is agency decision support. **Fix:** WP2 simulation-based inference (neural posterior estimation) with posterior-predictive checks, plus conformal prediction intervals on every dashboard-facing quantity.

W4. Learned controllers have no engineered safety.

DRL agents optimizing reward mixtures can and will crash; “safety” appears only inside reward terms and metrics. For deployment-adjacent claims, panels now expect constraints, not incentives. **Fix:** WP3 safety shields — friction-aware braking-envelope action filters (control-barrier-function style) that make hard violations impossible by construction, with shielded vs. unshielded ablations.

W5. Open-science promises contradict the proprietary platform.

The proposal pledges an open TransModeler–RL interface, open data, and open toolkits — but nothing is reproducible without a commercial license, and reviewers will say so. **Fix:** a dual-platform strategy (§5.6): develop at full fidelity in TransModeler, and release a validated open SUMO mirror of a benchmark subnetwork plus all learned models, making every paper’s results independently checkable.

W6. Hypotheses are not falsifiable as written; metrics are generic.

Thrust II’s central claims (“MARL can enable safe coexistence,” “cooperation will outperform selfishness”) carry no effect sizes, thresholds, or pre-registered metrics, and Thrust II’s evaluation is “cumulative reward.” **Fix:** the hypothesis table in §4 (H1–H5) with quantitative success criteria and ablation plans wired to each work package.

W7. Thrust integration is asserted, not mechanized.

The thrusts exchange “outputs” in prose, but no interfaces, data contracts, or feedback loops are specified — a classic “three projects stapled together” comment. **Fix:** the explicit dependency map in §5.6 (surrogate feeds control and complexity; the emulator guides scenario search; everything lands in one evaluated dashboard).

W8. Novelty is positioned on well-trodden applications.

Route guidance, GLOSA, and speed harmonization each have mature literatures (which the draft candidly reviews); a CISE panel discounts “we integrate known pieces” unless a crisp methods claim leads. **Fix:** lead with the methods — metro-scale world-model MARL, SBI-calibrated twins, learned complexity emulation, guided rare-event generation — and demote the three ITS applications to evaluation scenarios.

W9. Scope vs. staffing reads over-committed.

Two PhD students are asked to deliver a calibrated metro twin, novel MARL, complexity theory, open software, a dashboard, courses, and workshops. **Fix:** a third co-PI (me) with one additional student, and moving deployment-flavored deliverables (dashboard hardening, corridor pilots) onto USDOT vehicles (§6).

W10. Polish and compliance erosion.

Typos (“steet lights,” “Prenetration,” “reseach,” “collaboratino”), inconsistent terminology (SAC vs. CAV; Transmodeler vs. TransModeler), a duplicated “Phase 4” in Thrust II, a truncated reference ([23]), a misplaced citation ([64] on query substitution cited for surrogate modeling), and a Yann LeCun name-drop — individually small, collectively they cost benefit-of-the-doubt on everything else. **Fix:** the copy-edit punch list in Appendix A plus a terminology freeze (“CAV” throughout, defined once).

4. The repositioned project

Working title. *FROST-Twin: Foundations for Resilient Operations of Snow-affected Transportation — Learning-Accelerated Digital Twins for Safe Mixed-Autonomy Mobility.* (Alternate, if the team wants a wink at the ML crowd: *COLDSTART.*) The one-sentence pitch: *mixed-autonomy traffic in winter weather is a learning-enabled cyber-physical system whose safety, efficiency, and resilience phase behavior we will make predictable — with calibrated uncertainty — at the scale of a real metropolitan network.* Same twin, same partners, same complexity science; the scientific claims are now about learning-enabled CPS under environmental stress, which is exactly the CPS-FR remit inside Future CoRe (“intersection of computation, physical systems, and human interaction”).

4.1 Thrust architecture (old → new)

Thrust	Lead / co-lead	Keeps from the original	Adds (work packages, §5)
A. Twin & Inference	Banerjee / Chowdhury, Hasan	ATLAS + LiDAR + INDOT fusion; TransModeler build; V2V/V2I stack; GP calibration retained as baseline	WP2 simulation-based-inference calibration with full posteriors; conformal-UQ layer; open SUMO mirror of a benchmark subnetwork
B. Accelerated Safe Control	Chowdhury / Hasan	Three vehicle classes; MAPPO backbone; Stackelberg / Nash-bargaining modules at merges and intersections	WP1 graph world-model surrogate for fast training; WP3 safety shields with hard constraints; anchored validation ladder
C. Predictive Complexity & Resilience	Hasan / Chowdhury, Banerjee	Phase transitions, percolation thresholds, cascade metrics, synchronization analysis; agency-facing maps and dashboard	WP4 learned complexity emulator (minutes, not weeks, per what-if); WP5 generative stress-testing that hunts cascade-onset modes

4.2 Pre-registered hypotheses with quantitative success criteria

This table is the single highest-leverage addition to the resubmission: it converts aspirations into falsifiable, panel-checkable claims and gives every work package a finish line.

#	Hypothesis	Primary metric & threshold	WP
H1	A graph world model trained on at most ~20k simulator rollouts supports policy learning with at least 100× wall-clock speedup and at most 10% return degradation when policies transfer back to the full simulator.	Wall-clock ratio; surrogate-to-simulator return gap; 15/30/60-min rollout error (speed/density MAPE)	WP1
H2	Neural posterior estimation yields calibrated parameter posteriors and beats GP point calibration on unseen days, including snow regimes.	Posterior-predictive coverage within $\pm 5\%$ of nominal at 80/95%; at least 20% lower held-out-day RMSE vs. GP baseline	WP2
H3	Shielded cooperative MARL coexists safely with calibrated human drivers and beats both human-only and selfish-MARL baselines in heavy snow at moderate penetration.	Hard-safety violations = 0 by construction; near-miss rate (TTC under 2 s per 1k veh-km) reduced; network delay reduced by at least 15% vs. human-only at 20% penetration	WP3
H4	Percolation thresholds and cascade-size statistics are predictable from topology, demand, weather, and penetration features — and transfer to counties never seen in training.	Threshold relative error at most 10%; CRPS on cascade sizes; zero-shot county-transfer degradation at most 15%	WP4
H5	Guided generative scenario search discovers qualitatively distinct cascade-onset failure modes far faster than space-filling sampling at matched plausibility.	At least 5× distinct failure modes per 1,000 simulations vs. Latin-hypercube baseline; plausibility score vs. historical extremes	WP5

5. My proposed contribution: five work packages I can execute

Design constraints I imposed on myself: every package (i) runs on the standard PyTorch / graph-learning / RL stack, (ii) needs only two things from the existing plan — programmatic access to the twin through the team’s planned GISDK/Gym interface, and batch simulation capacity — and (iii) produces a standalone publishable result even if sister packages slip. Nothing depends on spiking networks, event cameras, or any speciality of my dissertation tooling; this is mainstream modern ML aimed at the proposal’s pressure points. Milestones assume a 4-year award.

WP1 — TwinBoost: graph world models as fast surrogates of the microsimulator

Motivation. Direct MARL inside TransModeler at metro scale is the proposal’s feasibility cliff (W1). Model-based RL solved the analogous problem elsewhere: learn the environment, train in the learned model, verify in the real one.

Approach. A lane-graph message-passing network with a recurrent/attention temporal head predicts link speeds, densities, and queue states at 30–60 s resolution, conditioned on signal states, boundary demand, weather covariates, and CAV penetration; optional physics-informed losses (flow conservation, cell-transmission consistency) regularize long rollouts. Training data: 5–20k stratified one-hour TransModeler rollouts batch-generated on HPC. The surrogate then serves as (a) the meso-level training environment for WP3 and (b) the ensemble engine behind WP4/WP5 sweeps.

Baselines / metrics. Per-link LSTM, DCRNN / Graph-WaveNet-class forecasters, and direct-simulator RL wall-clock. Metrics: multi-step rollout MAPE, queue-onset timing F1, interval calibration, and — decisively — the policy-transfer gap of H1.

Milestones / risks. M6 pilot on the I-465 NE quadrant; M12 metro scale; M24 fully in the WP3 loop. Risks: compounding rollout error (mitigate: scheduled sampling, shorter-horizon composition); distribution shift at high penetration (mitigate: DAgger-style active resampling from the full simulator).

WP2 — SBI-Twin: simulation-based inference for a calibrated, uncertainty-aware twin

Motivation. Agencies will act on this twin; point-calibrated parameters with no identifiability analysis (W3) are a liability. Simulation-based inference gives full posteriors over behavioral and environmental parameters at the cost of the same simulation budget the team already plans to spend.

Approach. Neural posterior estimation (sequential NPE via the open-source *sbi* toolkit) over grouped parameter vectors — car-following, lane-change aggressiveness, weather capacity multipliers, V2X latency/failure — conditioned on summary statistics of INDOT detector and probe data, stratified by weather regime; posterior-predictive checks on held-out days; a conformal-prediction wrapper turns every dashboard number into a calibrated interval. Priors on AV sensing range and detection degradation in snow are set from public winter AV datasets (CADC/Waterloo, Ithaca365/Cornell, Boreas/Toronto, WADS/Michigan Tech) instead of being assumed.

Baselines / metrics. The team's GP/Bayesian-optimization calibration (retained, and credited, as the baseline) and ABC rejection. Metrics: held-out-day RMSE on speeds/volumes/travel times; 80/95% interval coverage; identifiability diagnostics per parameter group (H2).

Milestones / risks. M9 clear-weather posterior; M15 snow regimes; M20 conformal layer live in the dashboard. Risk: high-dimensional posteriors (mitigate: sequential rounds + parameter grouping). Data governance: all inference runs on aggregate summaries, so raw INDOT feeds never leave their environment.

WP3 — ShieldedFlow: safety-constrained cooperative MARL for mixed autonomy

Motivation. Keep the team's MAPPO + game-theory design — it is sound — but make safety structural (W4) and training tractable (W1).

Approach. MAPPO backbone with density-dependent cooperative reward mixing; an analytic safety shield filters every action through friction-aware stopping-distance and time-headway envelopes (weather-dependent friction coefficient μ), in the spirit of control-barrier-function filters, so hard violations are impossible by construction. Stackelberg merge and Nash-bargaining intersection modules from the original Thrust II are layered at critical zones. Policies train primarily in the WP1 surrogate and are fine-tuned and evaluated in TransModeler.

Validation anchors. Before any new claims: reproduce the single-AV ring-road wave-dampening result (Stern et al. field experiment) and Flow-benchmark bottleneck findings as go/no-go gates — cheap, credible, and it converts W2 from a weakness into a selling point.

Baselines / metrics. Human-only calibrated baseline, selfish MARL, unshielded cooperative MARL, rule-based CACC. Metrics: violations (must be zero), near-miss rate, delay, throughput, jerk/comfort, fairness (Gini of user delays), and robustness under 10–50% V2X packet loss and latency jitter (H3). Milestones: M12 anchors; M18 corridor scale; M30 full simulation matrix.

WP4 — PercNet: a learned emulator of network complexity metrics

Motivation. Thrust III as written is post-hoc analysis of expensive simulation campaigns. Learning the map from (topology, demand, weather, penetration) to (percolation threshold, phase-transition location, cascade-size distribution) turns it into *resilience nowcasting* — the difference between a research finding and a planning tool.

Approach. A GNN / graph-transformer trained on WP1-accelerated ensembles predicts critical CAV penetration, the percolation curve, and cascade-size distribution parameters; attribution analysis (e.g., Shapley values, which Thrust II already planned) exposes which topological and demand features drive fragility, in planner language.

Baselines / metrics. Direct simulation (gold standard), classical structural predictors (betweenness, spectral gap). Metrics: threshold relative error, CRPS on cascade sizes, and — the headline — zero-shot transfer to held-out counties (H4). Milestones: M18 first emulator; M27 dashboard integration; M33 INDOT tabletop exercise using it live.

WP5 — StormForge: generative stress-testing of the twin

Motivation. Random scenario sampling wastes the simulation budget on benign days. The proposal already cites TeraSim-style adversarial scenario generation; this package brings that idea to the network scale, tied to winter weather.

Approach. A conditional generative model (diffusion or normalizing flow) over scenario tuples — spatiotemporal storm fields consistent with NCEI climatology, incident placements/durations fit to INDOT logs, demand surges — with rare-event guidance: samples are tilted toward high predicted cascade size from WP4 under an explicit plausibility constraint (likelihood bound), giving adversarial-but-plausible winters.

Baselines / metrics. Latin-hypercube and historical-replay sampling at matched compute. Metrics: distinct cascade-onset modes discovered per 1,000 simulations (target: at least 5× baseline), plausibility score, coverage of historical extreme days (H5). Milestones: M21 unconditional generator; M27 guided search; M33 red-team report delivered to INDOT.

5.6 Cross-cutting plan (fixes W2, W5, W7 in one page of the resubmission)

- **Compute.** Big Red 200 (IU's GPU-equipped HPC) for training; Jetstream2 — the NSF cloud resource housed at IU, allocated through ACCESS — for elastic CPU simulation farms running containerized TransModeler workers; a NAIRR Pilot allocation request as supplement. Named machines, estimated GPU-hours, and a scaling argument go in the facilities section; this alone neutralizes W1's reviewer math.
- **Open science without license contradictions.** Develop at fidelity in TransModeler; release (i) a validated open SUMO mirror of one benchmark subnetwork, (ii) the trained world model, policies, and emulator, and (iii) “IndyWinterFlow,” an anonymized detector-summary benchmark (subject to INDOT approval). Every paper becomes independently reproducible; the TransModeler–Gym interface still ships for licensed users.
- **Integration as data contracts.** One diagram in the resubmission: WP2 posteriors parameterize the twin; the twin generates WP1 training rollouts; WP1 accelerates WP3 policy learning and WP4/WP5 ensembles; WP4 guides WP5; all five report into a single evaluated dashboard reviewed annually by INDOT/IGIO. Each arrow is an API, not a sentence.
- **Validation ladder.** Tier 1: held-out-day replication at 0% penetration (their plan, kept). Tier 2: reproduction of published mixed-autonomy phenomena as gates. Tier 3: sensitivity and robustness analyses with WP2 uncertainty propagated end-to-end — conclusions reported as intervals, never points.

6. Where to submit: verified targets for the next six months

Deadlines below were checked against posted solicitations on August 13, 2026; items marked “verify” have live listings whose exact dates should be confirmed on the linked source (Appendix B) before we commit calendars. Note the Future CoRe participation cap: at most two proposals per person per rolling 12 months across all Future CoRe programs — so we choose our two shots deliberately.

Target	Vehicle / window	Size	Fit & what must change
1. NSF Future CoRe — CPS Foundations & Connected Communities (CPS-FR) — primary	NSF 25-543; accepted anytime; target dates Sep 10, 2026 and Feb 4, 2027 (recommended)	Up to \$1M / up to 4 yrs	Learning-enabled CPS under environmental stress: sensing (V2X/RSU) + learned control + human drivers in closed loop. Reframe per \$4; add hypothesis table, WP1–WP5, compute plan. Email cognizant CPS program directors a 1-page concept first.
2. Same solicitation — Robust Intelligence (RI) — alternate routing	NSF 25-543; same windows	Up to \$1M / up to 4 yrs	If the pitch leads with multi-agent learning + world models + UQ science and treats transportation as the testbed, RI is the cleaner panel. Decide after PD feedback; counts against the same 2-proposal cap.

Target	Vehicle / window	Size	Fit & what must change
3. FHWA Exploratory Advanced Research (EAR), FY2026	FY2026 solicitation on data-science applications in highway transportation — live listing; verify dates on SAM.gov	Contracts / cooperative agreements	Near-perfect domain fit for the twin + dashboard + INDOT partnership; deliverables re-angled to FHWA operations priorities (winter operations, incident management). The INDOT letter is gold here.
4. USDOT SMART, Stage 1 (Planning & Prototyping)	Applicant must be a public-sector entity — INDOT or the City of Indianapolis leads, IU as research partner; FY22–26 authorization (final year) — verify current NOFO	Planning-grant scale	The deployment-flavored slice (CAV Policy Evaluator, corridor pilots) belongs here, not in the NSF budget. Open the conversation with INDOT now; six weeks of agency lead time is typical.
5. NSF CIVIC Innovation Challenge (next cycle) — watch	Latest was NSF 24-534 (two-stage; Track A: climate/resilience); new cycle timing unannounced	\$75k Stage 1, up to \$1M Stage 2	Winter-storm mobility resilience with a civic partner as true co-lead. Keep warm with the MPO; move only if a solicitation posts.
6. CAIG resubmission — long shot, parked	NSF 25-530; last deadline Feb 4, 2026; no next cycle posted	2–3 senior personnel, up to 3 yrs	Viable only as a redesigned Earth-system project (AI for winter road-weather / mesoscale precipitation, twin as impact layer) with an atmospheric-science co-PI. Revisit only if the program reopens.

Also on the radar, for completeness. (a) *CRII is off the table for me*: the solicitation excludes PIs at Carnegie R1 institutions, and IU Indianapolis was designated R1 in February 2025 — worth knowing before anyone suggests it. (b) My *NSF CAREER* window opens July 2027 (outside this six-month plan), and the WP1/WP2 pilot doubles as its preliminary-results section — another reason to run the pilot now. (c) NSF announced *State and Regional AI Infrastructure Hubs* (NSF 26-513; consortium-scale, \$4–12M, November 4 deadline) on August 4, 2026 — not a vehicle for this project, but Luddy leadership should know Indiana could anchor a consortium, and our twin is exactly the kind of flagship use case such a hub cites.

7. Six-month roadmap (August 2026 → February 2027)

When	Actions	Owner
Aug 2026 (now)	Team alignment meeting; pick CPS-FR vs. RI framing; send 1-page concept to cognizant NSF program directors; request refreshed INDOT/IGIO letters; I draft \$4–\$5 as proposal text; submit ACCESS request for Jetstream2/Big Red 200 allocations.	All / SC
Sep 2026	Skip the Sep 10 target date unless PD feedback is unusually strong; instead launch the 10-week pilot: WP1 mini world model on the I-465 NE quadrant + WP2 clear-weather SBI on ~30 INDOT days; extend the INDOT data-use agreement to cover aggregate benchmark release; INDOT conversation on SMART Stage 1 lead.	SC (pilot), MAH/AB (agency)
Oct 2026	Pilot results v1 (speedup + calibration-coverage figures); internal red-team with two external readers; prepare FHWA EAR response per the posted BAA instructions.	SC / all
Nov 2026	Full 15-page draft: repositioned narrative, hypothesis table, WP1–WP5, pilot figures, compute + open-science plans; compliance and reference audit (Appendix A punch list).	All
Dec 2026	Revision round; letters finalized; facilities, data-management, and mentoring documents; budget shaped to the \$1M cap (\$8).	All

When	Actions	Owner
Jan 2027	Freeze and submit via Research.gov (Future CoRe accepts anytime) so the proposal is queued for the February panel cycle; EAR/SMART tracks proceed on their own clocks.	MAH (submitting PI)
Feb 4, 2027	Future CoRe target date — proposal in review, carrying pilot evidence no competing team is likely to have.	—

8. Proposed team structure, budget shape, and my role

- **Prof. Hasan (PI):** overall lead; complexity science (Thrust C lead); co-lead on multi-agent learning; submitting institution history (prior NSF award IIS-1909916 also keeps a future PFI-RP translation path open for the dashboard).
- **Prof. Banerjee (co-PI):** twin construction and GIS validation (Thrust A lead); TransModeler/GISDK interface; agency relationships (ATLAS, INDOT, IGIO); patent-backed network-reduction expertise feeding WP4 features.
- **Chowdhury (co-PI):** Thrust B lead and WP1–WP5 execution; one PhD student in deep learning / RL; modern training infrastructure (experiment tracking, containerized simulation workers, reproducibility discipline). I am asking for co-PI status, joint ownership of the methods papers, API access to the calibrated twin, and shared supervision credit — and I am offering to start the pilot before the ink is dry.

Budget shapes under the \$1M Future CoRe cap (indicative, not priced). *Shape A (lean):* two PhD students (Thrusts A and B), one summer month per investigator per year, workshops trimmed to one; Thrust C staffed by the Thrust B student in years 3–4 once WP1 amortizes simulation cost. *Shape B (preferred):* three PhD students (one per thrust), half a summer month per investigator, with the dashboard hardening, practitioner workshops, and corridor-pilot line items moved onto the FHWA EAR / SMART tracks where they are a better fit anyway. Either shape is honest about effort in a way the two-student original was not (W9).

9. Closing note

The CAIG outcome was a venue error, not a verdict on the science — and venue errors are the cheapest kind to fix. What the project needs to clear a CISE panel is exactly what I do: make the learning tractable (WP1), the calibration honest (WP2), the control safe (WP3), the complexity predictive (WP4), and the stress-testing sharp (WP5), all stated as falsifiable hypotheses with numbers attached. If the team will have me as co-PI, I will have the I-465 pilot running within two weeks of our first meeting, and pilot figures in hand before the November draft. I would welcome the chance to walk through this document together. — S.C.

Appendix A. Copy-edit and compliance punch list for the existing draft

Concrete items found on a close read; fifteen minutes of fixes that buy disproportionate reviewer goodwill.

- “steet lights” → “street lights” (§1.2 Thrust I and §2); “Prenetration” → “Penetration” (Thrust I heading).
- “reseach” → “research” (Thrust III overview and §1.2); “collaboratino letter” → “collaboration letter” (§5); “deleop” → “develop” (§2.2); “Transportatino” → “Transportation” (§2.1); “followoing” → “following” (§4.2).
- Grammar in the Project Summary: “The project will uses novel economics...” and “Primary innovation is an CAV-ready digital twin”; §1.4 “The project is lead by” → “led by”; §10 “Co-PI Banerjee do not have” → “does not have.”
- Terminology freeze: the Overview uses “SAC vehicles,” the rest “CAV” — define “CAV” once and use it throughout; standardize “TransModeler” capitalization (currently mixed with “Transmodeler”).
- Thrust II research plan announces four phases, then narrates “third phase,” “final phase,” and a second “Phase 4” — renumber and reconcile.

- Reference [23] (Goldman Sachs) is truncated mid-sentence (“forecast to comprise 10 URL...”); repair or replace.
- Reference [64] (Jones et al., “Generating query substitutions,” WWW 2006) is cited as support for surrogate-model training — almost certainly a bibliography paste error; replace with an actual surrogate-modeling citation.
- Remove the “Yann LeCun’s vision of physical AI” sentence (§1.3); make the physical-grounding argument on its own merits — name-drops read poorly in panel.
- Evaluator name appears as both “Dr. Steven Aldrich” and “Dr. Aldrich” (§1.4) — pick one spelling.
- Figure 1 reads as a composited raster; replace with an original vector diagram (panels increasingly comment on AI-generated-looking figures), and fix its caption (“interacting with AVs”).

Appendix B. Sources and starting points (verified Aug 13, 2026)

NSF 25-543 — Future Computing Research (Future CoRe): target dates, \$1M/4-yr project class, 2-proposal cap
[nsf.gov/funding/opportunities/future-core-computer-information-science-engineering-future-computing/nsf25-543/solicitation](https://www.nsf.gov/funding/opportunities/future-core-computer-information-science-engineering-future-computing/nsf25-543/solicitation)

CPS Foundations & Connected Communities program page (CPS-FR / CPS-CIR tracks)
[nsf.gov/funding/opportunities/cps-cyber-physical-system-foundations-connected-communities](https://www.nsf.gov/funding/opportunities/cps-cyber-physical-system-foundations-connected-communities)

NSF 25-530 — CAIG solicitation (scope: “scientific understanding of the Earth system”; last deadline Feb 4, 2026)
[nsf.gov/funding/opportunities/caig-collaborations-artificial-intelligence-geosciences/nsf25-530/solicitation](https://www.nsf.gov/funding/opportunities/caig-collaborations-artificial-intelligence-geosciences/nsf25-530/solicitation)

FHWA Exploratory Advanced Research program (FY2026 data-science solicitation — confirm dates on SAM.gov)
highways.dot.gov/research/research-programs/exploratory-advanced-research

USDOT SMART grants program (Stage 1 planning grants; public-sector applicants)
transportation.gov/grants/SMART

NSF CIVIC Innovation Challenge (latest solicitation NSF 24-534; watch for next cycle)
[nsf.gov/funding/opportunities/civic-civic-innovation-challenge](https://www.nsf.gov/funding/opportunities/civic-civic-innovation-challenge)

Jetstream2 (NSF cloud housed at IU; allocations via ACCESS) and IU Big Red 200 — facilities for the compute plan
jetstream-cloud.org · access-ci.org

NAIRR Pilot — national AI compute allocations
nairrpilot.org

sbi toolkit for simulation-based inference (WP2)
github.com/sbi-dev/sbi

Flow project — mixed-autonomy RL benchmarks used as WP3 validation anchors
flow-project.github.io

Winter AV datasets for WP2 priors: CADC (Waterloo), Ithaca365 (Cornell), Boreas (Toronto), WADS (Michigan Tech)
cadcd.uwaterloo.ca · ithaca365.cs.cornell.edu · boreas.utias.utoronto.ca

Program deadlines and structures changed substantially during the 2025–26 NSF reorganization; every date above was re-verified against the posted solicitation text on August 13, 2026, but confirm on the official page before committing internal deadlines.